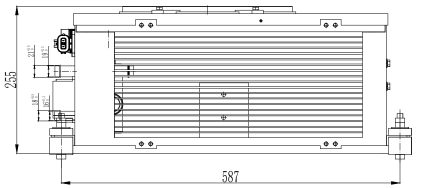
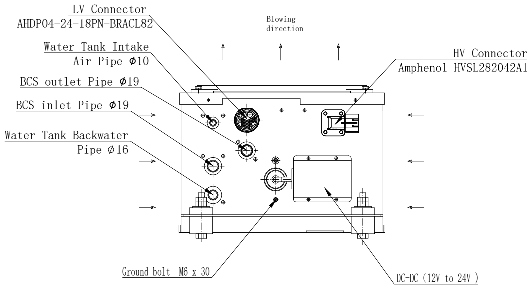
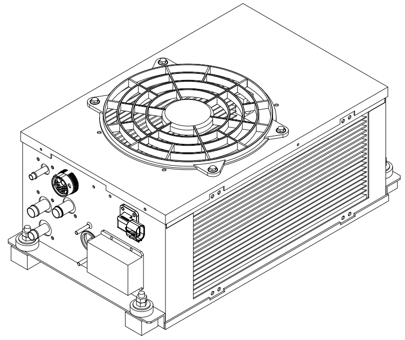
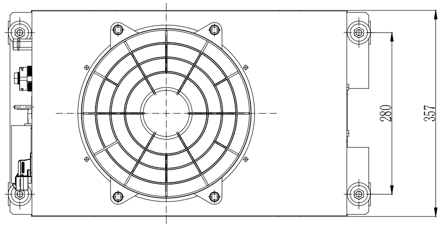
CBAF010009

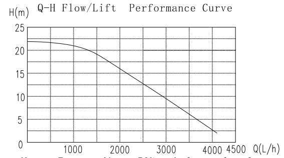








Built-in pump performance curve

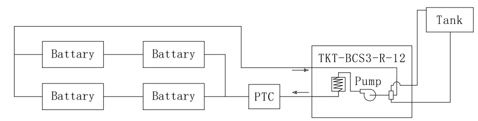


Note: Test medium: 50% ethylene glycol antifreeze

Specification

Model NO.	TKT-BCS3-R-12	Expansion Valve	SH207-1
Cooling Capacity	3KW	Pump Power (V/A)+ Cond Fan Power (V/A)	DC12V/30A
Heating Capacity	5KW		
HV Power Supply	DC220-450V	Control Power (V/A)	DC12V/5A
HV Rated Input Power	1.7KW(Cool)/5KW(Heat)	Condition	AMBIENT TEMP:40℃
Refrigerant	R134a (350+20)g		WATER OUT TEMP:10-15℃
Weight	25KG		WATER FLOW: 45-60L/min(10-6M H_2O)

Recommended typical flow chart



Note:PTC and Tank are arranged outside the BCS.

Connectors Details for BTMS of TKT-BCS3-R-12 (320V/12V)

Serial No.	Interface Function	BTMS Side	Vehicle Side	Line Position	Wire Diameter (mm²)/Color	Supporting Fuse and Description	Jacketed Butt Interface picture
1	BTMS to vehicle CAN communication	Amphenol AHDP04-24-18PN-BRACL82	Amphenol AHDP06-24-18SN-BRACL82	1-CAN-H 2-CAN-L	1-0.5/Yellow 2-0.5/Green	Connect vehicle CAN communication	
	BTMS water-replenishing tank, water level normally closed switch interface			3-WL 4-Negative	3-1.0/Brown and White 4-1.0/Black	Connected to the normally closed water level switch of the expansion tank (inverted)	
	BTMS control power			5-Negative 6-Positive	5-1.0/Blue 6-1.0/Red	Connect to vehicle DC12V power supply negative Connect the vehicle key control switch DC24V with 5A-10A fuse positive	
	DC12V main power supply for BTMS			A-Positive B-Negative	A-6.0/Red B-6.0/Black	Connect the vehicle DC12V with 50-60A slow-melt fuse main power supply positive Connect to vehicle DC12V main power supply negative	
	BTMS to PTC CAN communication			13-CAN-H 14-CAN-L	13-0.5/Yellow 14-0.5/Green	Connected to PTC heater CAN communication	
	PTC control power			15-Negative 16-Positive	15-0.75/Black 16-0.75/Red	Connected to the negative pole of the PTC heater DC24V power supply Connected to the positive pole of the PTC heater DC24V power supply	
2	High voltage power supply	Amphenol HVSL282042A1	Amphenol HVSL282062A104I	1-Positive 2-Negative	1-4.0/Red 2-4.0/Black	Connect the vehicle DC320V with 25-30A fast-melt fuse main power supply positive Connect to vehicle DC320V main power supply negative	
3	BTMS internal control debug interface	AMP 282106-1	AMP 282088-1	1-CAN-H 2-CAN-L	1-0.5/Yellow 2-0.5/Green	Connect to internal debugging controller CAN communication	 For Internal Testing Only
				3-Negative 4-Positive	3-Black 1.0 4-Red 1.0	DC24V power supply is negative DC24V power supply is positive	

Technical requirements:

1、For product layout, please choose a location without direct rainwater, the space requirements are convenient for installation and maintenance, the air inlet and outlet of the fan are not blocked, and there is no short circuit in the return flow of the wind;

2、It is recommended to use 50% ethylene glycol solution as the coolant, and the freezing point temperature is lower than -15℃ and not higher than (minimum ambient temperature in winter -5)℃;

3、The product control power supply is controlled by the vehicle key switch, High and low voltage interface definitions see the electrical schematic diagram for details;

4、The product is not equipped with manual manipulator and accepts the CAN communication control of the vehicle to run automatically. The control logic is detailed in the CAN communication protocol.

TKT-BCS3-R-12	CBAF010009	A		
Cooling+Heating	BTMS Dimension Drawing			
Design	Eric			
Proofread	Yanlin Zhu			
Check				
Approve	Wuxing He	1ST	ANGLE	
C	S	D	P	
Total	Page	No.	Page	

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